Signatures of sexually antagonistic intragenomic coevolution in *Drosophila melanogaster*

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Abstract

In species with separate sexes, it is sometimes a disadvantage to share the same genome. Selection within each sex will act to optimize the genome in that sex's favour, potentially leading to intergenomic conflict between the sexes. During a long-running evolution experiment (female-limited X chromosome, or FLX), we found evidence suggesting that different parts of the genome were evolving towards different sex-specific optima, suggesting the presence of an intragenomic conflict as well. To test this hypothesis, we used a sophisticated crossing scheme where we attempted to disentangle responses to selection on the sex chromosomes and autosomes by swapping them between the selection regime (FLX) and the control wildtype (Cwt), and between the methodological control regime (CFM) and Cwt. We then assayed three phenotypes and analysed expression data for the four new genotypes. Our *a priori* expectation was that sex chromosomes from FLX should show evidence of feminization, but that autosomes from both FLX and CFM should show evidence of masculinization. Our expectations were partially fulfilled; however, we also found some unexpected results. Strikingly, a mismatch between the sex chromosomes and autosomes seemed to have an order of magnitude greater effect on female expression profiles compared to males.

1. Introduction

Sexual conflict arises when males and females have different reproductive strategies. It can manifest in multiple traits and behaviours differentiating between the sexes when they take different routes to achieve their reproductive optima [1]. Sexual conflict is usually considered to be composed of two more or less distinct categories, inter- and intralocus conflict. Interlocus conflict is typically considered to be the result of two different loci being selected antagonistically in each sex. In contrast, if the same locus is subject to antagonistic selection between males and females, this is intralocus sexual conflict, also sometimes known as sexual antagonism [2]. Sex-specific selection on males and females can therefore turn the genome into a battleground over fixation of male-benefit versus female-benefit loci (or alleles). This may have implications for coevolution and conflict within the genome. A well-established example of potential intragenomic conflict is between the mitochondrial and nuclear genomes. This has been support by both theoretical models and empirical data suggesting that asymmetric selection pressures and inheritance between the sexes can lead to such conflict [3]. Even within the nuclear genome there is evidence of conflict due to sex-specific selection pressures due to the asymmetric inheritance of the sex chromosomes [4]. Even though autosomes are inherited equally between the sexes, a recent model by Muralidhar and Coop [5] showed that autosomes can also respond in a sex-specific way to sexually antagonistic selection. They found that because X chromosomes are more often inherited via females compared to the autosomes, the X should exhibit a female-biased response to selection and the autosomes a relatively male-biased response to selection (at least in cases where X chromosomes experience stronger or more prevalent female-specific selection, for example in species that lack dosage compensation).

We have previously carried out an evolution experiment in *Drosophila melanogaster* to examine what happens if the X chromosome is only exposed to selection in females (i.e. female-limited X-chromosome evolution, or FLX – see methods for details). We assumed that traits under sexual conflict would be released from male-specific selection and move towards the female optimum under this selection regime. We specifically focused on the X chromosome because its asymmetric inheritance pattern between the sexes could have implications for sexual conflict. On one hand female-biased inheritance of the X chromosome would point to the X chromosome being feminized relative to the autosomes, however on the other hand faster-X effects resulting from hemizygosity in males would instead point to the X chromosomes being masculinized relative to the autosomes [6,7]. Female-limited inheritance of the X in the FLX experiment was enforced using a balancer chromosome (FM7a, [8]), so in addition to the FLX selection regime, we included a methodological control regime which was designed to control for effects of adaptation to the balancer (denoted CFM), as well as a wildtype control regime which was designed to control for other aspects of the experimental protocol such as isolation of flies as virgins and reduced population size (denoted Cwt, see methods for details). After 40 generations of the FLX evolution experiment it became clear that the FM balancer seemed to have impacted the flies in ways we had not foreseen, as the CFM regime continuously defied our expectations [9–11]. Follow-up experiments revealed that males carrying the FM balancer performed extremely poorly when in competition with wildtype males, presumably due to the phenotypic markers on the FM which affect eye-colour and shape, making the FM males functionally blind (Lund-Hansen unpublished data). This led us to speculate that the presence of the balancer caused a shift in the intensity of interlocus and/or intralocus sexual conflict within our selection regimes [10,11]. The design of our experimental protocol made it likely that any adaptation to compensate for deleterious effects of the balancer in males would occur primarily via the autosomes.

We therefore wanted to test if we unintentionally had caused an intragenomic conflict between the sex chromosomes and autosomes due to our experimental set up. To disentangle adaptation on the X chromosome from adaptation on the autosomes, we introgressed FLX and CFM sex chromosomes into a wildtype autosomal background, and we introgressed wildtype sex chromosomes into a FLX or CFM autosomal background (Figure *1* & S1). After the target genotypes were produced, we collected expression data and measured three phenotypic traits we *a priori* suspected to show evidence of conflicting selection pressures: locomotory activity, sperm offence and sperm defence [9,10]. These traits seemed to be consistent with masculinization in the CFM regime (presumably mainly via autosomal adaptation) which was countered by feminization in the FLX regime (presumably mainly via adaptation on the X chromosome). In addition, there was also evidence that the genetic architecture of locomotory activity had changed in the FLX selection lines (Manat et al. submitted), making it a particularly interesting trait to study in the context of selective responses on the sex chromosomes versus autosomes.

Our expectations were that one or both of the FLX and CFM autosome genotypes should show evidence of adaptation for male fitness compared to the FLX and CFM sex chromosome genotypes (i.e. result in masculinization of expression, increased locomotory activity, and increased performance in sperm competition), while the FLX sex chromosome genotype should show evidence of adaptation for female fitness compared to the CFM sex chromosome genotypes (i.e. result in feminization of expression, decreased locomotory activity, and decreased performance in sperm competition), similar to results from Muralidhar & Coop [5]. Our results were partly consistent with these predictions. Expression data showed some signatures of feminization in FLX and masculinization in CFM, but these differences were not clearly restricted to the autosomes or sex chromosomes. We also found evidence of feminization of locomotory activity in females carrying FLX sex chromosomes, but there were no differences between genotypes in male performance in sperm competition. Unexpectedly, we found an order of magnitude more differentially expressed genes in females compared to males, suggesting that mismatches between X chromosomes and autosomes have much more wide-ranging effects in females than in males [4].

2. Methods

(a) *Drosophila melanogaster* and the FLX evolution experiment

The flies used for the crosses in this paper were derived from an evolution experiment started in 2013 [9]. Lines were started from the wildtype population LH_M and all flies were kept under the standard LH_M protocol: 25°C, 12/12 light/dark, 60% relative humidity, and fed on a cornmeal-molasses-yeast medium [12]. The evolution experiment consisted of three regimes; one selection regime and two control regimes, with four replicate populations in each regime. In the female-limited X chromosome (FLX) selection regime we enforced the inheritance of the X chromosome through the matriline line by using an X chromosome balancer (FM7a: In(1)$B^1/sc^8/v^{Of}/w^a/y^{31d}$). To control for any unexpected effects of adaptation to the FM balancer, we included a control regime (CFM) where the X chromosome spent every third generation in males. Lastly, we had a wildtype control regime (Cwt) to control for any other effects of the experimental protocol, such as reduced population size and collecting the females as virgins. For a full description of the experimental protocol, see [9].

(c) Creation of target genotypes

We created four novel genotypes in this experiment: FLX sex chromosomes with Cwt autosomes (hereafter denoted FLXsex), CFM sex chromosomes with Cwt autosomes (CFMsex), Cwt sex chromosomes with FLX autosomes (FLXauto), and Cwt sex chromosomes with CFM autosomes (CFMauto) (Figure *1*). Crosses to produce these target genotypes were started at generation 169 of the FLX evolution experiment and required six rounds of crosses to complete using the "clone generator" system developed by Rice [13]. The crosses are described in detail in the supplementary material, S1.

(d) Sperm competition assay

As soon as the target genotypes were produced, we carried out a sperm competition assay. We measured both sperm defence and sperm offence. For sperm defence we measure the ability of the target males' sperm to resist replacement by a second male's sperm. This is done by first mating the target male with a female carrying a recessive brown-eye marker and mating the female with a male that also carries the brown-eye marker. All offspring sired by the target males will therefore be red-eyed and the proportion of red-eyed offspring is used as a measurement of the success of the sperm. For sperm offence we measure the ability of the target males' sperm to replace the sperm of another male. The assay is done in the same way as sperm defence, though the target males are now the second male to mate. Both assays were done in two blocks with 20 males per genotype per replicate population (although a few data points were usually excluded from each set due to loss of flies, inability to confirm first or second mating, etc.). For full description of the assays see supplementary material, S2.

(i) Statistical analysis

Performance in sperm competition was analysed using generalized linear mixed models in R (R version 4.5.2 [14]). Because the data showed evidence of zero-inflation for both sperm defence and sperm offence, we set up models that controlled for zero-inflation using the glmmTMB package [15,16]. Genotype was treated as a fixed effect, and Replicate population was nested within Genotype as a random effect in order to avoid pseudoreplication. Data was collected over two blocks, so block was included as a fixed effect to control for differences between blocks. Note that because block is a nuisance variable, this effect could potentially be treated as a random effect instead, but results were qualitatively similar regardless of whether block was treated as a fixed or random effect, and we judged it to be more appropriate as a fixed effect since it had so few levels, and because this provided better model convergence. Model assumptions were tested using diagnostics from the DHARMa package [17]. We checked for evidence that zero-inflation was non-randomly distributed across genotypes, blocks, and replicates by defining ziformula = ~Genotype+(1|Block)+(1|Genotype:Rep), but there was no evidence of any significant effects other than for the intercept. The AIC value for the simplified model (ziformula = ~1) was lower than for the full zero inflation model (ziformula = ~Genotype+(1|Block)+(1|Genotype:Rep)) for both sperm offence and sperm defence, so we only report the results from the reduced model here, which was formulated as follows: glmmTMB(cbind(Red, Brown)~Population+(1|Block)+(1|Population:Rep), ziformula = ~1, data = defence/offence, family = binomial). Posthoc tests of pairwise differences between genotypes were carried out using Tukey tests from the emmeans package [18].

(e) Locomotion assay

One generation after the target genotypes were generated, we measured locomotory activity in both sexes. We observed one fly alone in a vial and noted if it was walking around or not. Each vial was observed 10 times and all vials were observed within an observation window which was usually no more than 60 minutes. This assay was carried out over two days and repeated at different times of the day. In total 16 flies were observed per sex, genotype, and replicate population.

(i) Statistical analysis

Because data was collected on two different days and at different times of the day, we included effects of day (as a fixed effect since it only had two levels) and time (as a random effect since it had nine levels) to control for age and circadian rhythm effects. As with the analysis of the sperm datasets, Genotype was included as a fixed effect and Replicate population as a random effect nested within Genotype. Males and females were analysed within the same model, since we were interested to know if there were any differences in how male and female locomotion were influenced by Genotype, so Sex and the Sex*Genotype interaction were also included as fixed effects. However, there was evidence of overdispersion resulting from zero-inflation, so as for the sperm data we used glmmTMB to control for zero-inflation. We also checked for evidence of non-random distribution of zero-inflated data but only found weak evidence of this (a significant effect of an increased number of zeros on day 2), and AIC values suggested that the simplified model (with only an intercept effect for zero inflation) was a better fit overall. We therefore report results from this model formulation: glmmTMB(cbind(active, passive)~Genotype*Sex+Day+(1|Genotype:Rep)+(1|Time), ziformula = ~1, data = locomotion, family = binomial).

Posthoc tests of pairwise differences between genotypes were carried out using Tukey tests from the emmeans package. However, carrying out pairwise tests between all genotypes in both sexes resulted in a large number of comparisons requiring correction for multiple testing. Since we were most interested to know if the pattern of differences was consistent between the sexes or not, rather than doing direct comparisons between males and females, we only carried out posthoc comparisons between genotypes within each sex: emmeans(model, list(pairwise ~ Genotype | Sex), adjust="tukey").

(f) RNA extractions

Nine generations after the genotypes were created, three biological replicate pools of 20 flies per genotype, sex, and replicate population were collected for RNA extraction (for a total of 96 samples). We did not collect samples from pure Cwt genotypes since we were primarily interested in comparisons between target genotypes and wanted to restrict the total number of possible posthoc comparisons. Flies were collected on day 12 after oviposition and left in a vial with food in same-sex and same-genotype groups for two days. The flies were then quickly anaesthetised using CO_2, divided into pools of 20 flies, and immediately snap-frozen in liquid N_2 before being stored at -80°C. RNA extractions were carried out using the RNeasy plant mini kit (Qiagen) according to the manufacturer's instructions. The samples were sent to National

Genomics Infrastructure (NGI) Sweden for library preparation (Illumina TruSeq Stranded mRNA (polyA)) and sequenced on a NovaSeq6000. For each sample 500ng of total RNA was used, and each sample was run on three sequencing lanes which achieved 27-128M reads per sample.

(g) Bioinformatics

Because each sample was run on three different lanes, we merged the three samples before the next steps. Following the protocol in [19] we used trimmomatic/0.39 [20] to trim the files, hisat2/2.2.1 [21] for alignment to the *Drosophila melanogaster* genome version dmel_r6.65 [22], and subread/2.1.1 [23] to create the count table. Preliminary analysis revealed substantial differences in expression profile between males and females (S5), so we decided to analyse the sexes separately. Differential expression between the novel genotypes were analysed using DESeq2 [24], using the formula: design = ~ Genotype + Genotype:nested.Rep.

(i) Downstream analysis

We carried out all six pairwise comparisons for the four novel genotypes. For six comparisons there are 63 possible significant intersections and one complement with no significant transcripts (2^6) (this is also the reason why we do not include data for full "pure" FLX, CFM, and Cwt genotypes). In order to determine if the total number of transcripts in each intersection was similar between the sexes, we carried out a Spearman correlation test. For downstream analyses we selected the seven most numerous intersections in females and the corresponding intersections in males, and followed the methods in [25] (see supplementary information for more details, S3). These included:

- Chi-squared or Fisher exact tests (depending on the number of transcripts) to determine if transcripts were previously characterized as (1) sex-biased [26], (2) associated with sexual conflict [27,28], (3) associated with sexual antagonism [29,30], (4) associated with tissue-specific expression [31], or (5) mito-nuclear conflict [32–34] more often than expected by chance.
- Mean-rank gene set enrichment Wilcox tests to determine if changes in expression in our transcripts were significantly associated with expression differences previously characterized as associated with sex-specific fitness [29].
- Analysis of overrepresented gene ontology (GO) terms using the package *clusterProfile* [35] for enriched GO terms.

- Chi-squared or Fisher exact tests (depending on the number of transcripts) to determine if transcripts were non-randomly located within the genome (we were specifically interested to check for over- or underrepresentation on the X chromosome compared to the autosomes).

3. Results

(a) Phenotypic experiments

(i) Sperm competition assay

Genotype was not significant for either sperm defence (Wald's χ^2 = 0.867, df = 3, P = 0.8335) or sperm offence (Wald's χ^2 = 6.08, df = 3, P = 0.1077). See figures (S4) and table S5A&B for full model results.

(ii) Locomotion assay

There were significant effects of Sex (Wald's χ^2 = 5.83, df = 1, P = 0.016) and Sex*Genotype (Wald's χ^2 = 22.02, df = 3, P = 6.448×10^{-5}) on locomotory activity (see Figure 2 and S5C). Males had slightly lower activity overall, which is somewhat unexpected, but may be related to effects of coevolution with the FM balancer and/or experimental design, since we have previously found higher overall activity in males from our experimental evolution lines when assaying this phenotype in groups [9]. Posthoc comparisons revealed a significant difference between females carrying CFM X-chromosomes compared to FLX X-chromosomes (P = 0.018, see Figure 2, S5D), consistent with our *a priori* expectation that FLX X-chromosomes should decrease locomotory activity compared to CFM X-chromosomes, as well as a significant difference between females carrying CFM autosomes compared to FLX X-chromosomes (P = 0.037, see Figure 2, S5D), which is partially consistent with our expectation that the autosomal genotypes should be relatively masculinized. However, there were no significant differences among male genotypes.

(b) Gene expression

We found 16785 transcripts with at least one read. 5725 transcripts (34.11%) were significantly differentially expressed between the genotypes in females (S7), and 588 in males (3.50%, S8). Of the 63 possible intersections between the six pairwise significant comparison sets, we found at least one differentially expressed transcript in 47 of the intersections in the female data set, and 40 in the male data set (Figure 3A&B), most of which (36) were shared between the sexes.

The distribution of transcripts across these intersections was very uneven (Figure 3), but although there were many fewer significant transcripts in males we found that the number of transcripts in each intersect was significantly correlated between the sexes (Spearman correlation, $P = 1.255*10^{-6}$). Because there were so many unique intersections with at least one significant transcript, we decided to focus on the seven most numerous intersections in females and their corresponding partners in the male data set. We label these seven intersections groups A-G in decreasing numerical order (Figure 3A) and will use this notation throughout from this point forward (Table 1 and Figure 4). These seven groups account for almost 80% of all significant transcripts in the females and approximately 50% in males. As about one quarter of the transcripts in the females were up-regulated and three quarters were down-regulated, we also decided to analyse the up- and down-regulated transcripts separately within each group.

Females had a highly significant overrepresentation of up-regulated female-biased transcripts and an underrepresentation of down-regulated female-biased transcripts in all groups (S9), and the reverse for male-biased transcripts (i.e. underrepresentation among upregulated transcripts and overrepresentation among downregulated transcripts). Upregulation in our comparisons generally means higher expression in the following order FLXsex – FLXauto – CFMsex – CFMauto, so this suggests higher expression of significant transcripts in FLX females compared to CFM females overall. Patterns of sex bias were less consistent in males (evidence of underrepresentation of male-biased transcripts and overrepresentation of unbiased transcripts among downregulated genes in groups B and C; S10).

We also looked at significant intersections that were only found in one sex (Female: 11, Male: 4), to see if these seemed to be unusual in some way. Somewhat surprisingly, we found underrepresentation of sex biased transcripts in female-specific groups compared to the set of all significant transcripts, but no significant effects in the male-specific groups (S11 & S12). This suggests that significant transcripts in the female-specific groups are primarily due to differences in power of detection between the sexes (since there were more differentially expressed transcripts in females than in males) and are not related to overall differences in sex-specific expression profile. This is consistent with the other downstream analyses, which generally revealed fewer significant effects in males (see below). Results for each group are detailed below; due to the large total number of analyses, non-significant results are not explicitly listed in the main text but can be found in the supplementary information.

(i) Group A

In group A, we found 2150 significant transcripts in females and 20 transcripts in the corresponding group in males (Table 1). Characteristic for this group is that CFMauto is significantly different from the three other new genotypes (see Figure 4A for a schematic representation). Female transcripts with higher expression in CFMauto compared to the other genotypes (2021/2050) were significantly positively associated with male fitness (S13), and enriched for genes that are associated with sexual conflict in *D. melanogaster* and *D. pseudoobscura* (S15), genes whose expression is associated with male-benefit sexual antagonism (S17B), and mitosensitive genes (S43), as well as genes associated with expression in 10 different tissues (S19). Common themes among GO terms for these transcripts were behaviours, response to stimuli, development, and neuron function (S33). These results are all generally consistent with masculinization on CFM autosomes. Female transcripts with lower expression in CFMauto compared to the other genotypes (129/2150) were enriched for genes that are associated with sexual conflict in *D. melanogaster* (S15), for genes associated with expression in ovaries (S19), and for X-linked genes (S46). These results are generally consistent with feminization in FLX genotypes (and CFM sex chromosomes) compared to CFM autosomes.

(ii) Group B

We found 952 significant transcripts in group B in females and 112 in males, which is the most numerous male intersection (Table 1). Characteristic for this group is that CFMauto is significantly different from CFMsex (Figure 4B). Female transcripts with higher expression in CFMauto compared to CFMsex (382/952) were positively associated with female fitness and female-benefit sexual antagonism (S13), and enriched for genes associated with sexual conflict in *D. melanogaster* (S15; this pattern was also found in males, see S16A), genes whose expression is associated with sexual antagonism (S17), and mito-annotated and mito-proteome genes (S43, overrepresentation of mito-proteome genes was also found in males, see S44), as well as genes associated with expression in 10 different tissues (S21; two tissues, heart and midgut, were also significant in males, see S22). Consistent with this, GO terms for these transcripts were often related to mitochondria and mitochondrial function (S34). Female transcripts with higher expression in CFMsex compared to CFMauto (570/952) were negatively associated with female fitness, positively associated with male fitness, and positively associated with male-benefit sexual antagonism (S13). They were also enriched for genes associated with sexual conflict in *D. pseudoobscura* (S15B), genes associated with expression in crop and ovaries (S21), and mito-proteome genes (S43). GO terms for these transcripts were again associated with mitochondria and mitochondrial function, as well as RNA translation (S34; and antioxidize activity in males; S35). Results from this group are generally consistent with masculinization of CFM sex chromosomes compared to CFM autosomes, which was a pattern we did not predict, but which somehow seems to be associated with selection on metabolism. However, we note that this group is also somewhat consistent with our *a priori* prediction of masculinization in CFM sex chromosomes (although we expected the main difference to be compared to FLX sex chromosomes rather than CFM autosomes).

(iii) Group C

We found 549 significant transcripts in group C in females and 64 in males (Table 1). Characteristic for this group is that FLXauto is significantly different from CFMauto (Figure 4C). Female transcripts with higher expression in CFMauto compared to FLXauto (372/549) were significantly enriched for genes associated with sexual conflict in *D. melanogaster* (S15, this pattern was also evident in males; S16), as well as for genes associated with 11 tissues (S23; and with midgut in males, see S24). GO terms for this set of transcripts were associated with receptor activity (S36). Female transcripts with higher expression in FLXauto compared to CFMauto (177/549) were negatively associated with female fitness and positively associated with male fitness and male-benefit sexual antagonism (S13). They were also enriched for genes associated with expression in the ovaries and the salivary gland (S23; as well as the thoracicoabdominal ganglion in males, see S24), and mito-proteome genes (S43). GO terms for this set of transcripts were related to the endoplasmic reticulum and RNA translation. Results from this group are generally consistent with masculinization of FLX autosomes compared to CFM autosomes, which again was not a pattern which we expected *a priori*. However, this pattern could potentially be explained as a result of autosomal compensation for female-limited selection of the X-chromosome, similar to results from Muralidhar & Coop [5].

(iv) Group D

There were 475 significant transcripts in group D in females and 30 in males (Table 1). Characteristic for this group is that CFMauto is significantly different from CFMsex and FLXauto (Figure 4D). Female transcripts with higher expression in CFMauto compared to the others (284/475) were positively associated with female fitness and female-benefit sexual antagonism (S13; the same was also true in males, see S14). They were also enriched for genes whose expression is associated with sexual antagonism (S17), as well as genes associated with expression in nine different tissues (S25; also the midgut in males, see S26). There was one significant GO term for these transcripts in males, related to peptidase activity (S38). Female transcripts with lower expression in CFMauto compared to the other genotypes (191/475) were negatively associated with female fitness (S13), and significantly enriched for genes associated with sexual conflict in *D. pseudoobacura* (S15), genes associated with expression in the ovaries and malpighian tubules (S25), mito-annotated and mito-proteome genes (S43), and had GO terms primarily related to the mitochondria (S37). This set of transcripts also showed underrepresentation of X-linked genes (S45). Although the pattern of significant differences in this group is similar to group A, the properties of the transcripts in the group suggest that is capturing something different, since they are most consistent with feminization of the CFM autosomes rather than masculinization. Again, this seems to potentially be associated with mitochondrial/mitonuclear effects and could be explained by autosomal compensation for deleterious interactions between the FM balancer and the mitochondrial genome.

(v) Group E

There were 362 significant transcripts in group E in females and 43 in males (Table 1). Characteristic for this group is that FLXsex is significantly different from CFMauto (Figure 4**Error! Reference source not found.**E). Female transcripts with higher expression in CFMauto compared to FLXsex (264/362) were negatively associated with female fitness (S13), and enriched for genes associated with expression in seven tissues (S27). Male transcripts with higher expression in CFMauto compared to FLX (26/43) were enriched for genes associated with sexual conflict in *D. melanogaster* (S16). Female transcripts with higher expression in FLXsex compared to CFMauto were also enriched for genes associated with sexual conflict in *D. melanogaster* (S15), and enriched for genes associated with expression in the heart and midgut (S27). There was only one significant GO term for this set of transcripts, related to regulation of anatomical structures (S39). The pattern of differences in this group is likely related to overall differences between the selection regimes.

(vi) Group F

We found 304 significant transcripts in group F in females and 5 in males (Table 1). Characteristic for this group is that FLXsex and FLXauto are significantly different from CFMauto (Figure 4F). Female transcripts with higher expression in CFMauto compared to the others (264/304) were enriched for genes associated with sexual conflict in *D. melanogaster*

(S15), and genes associated with sexual antagonism (S17), as well as genes associated with expression in nine tissues (S29). Three GO terms were associated with this set of transcripts; G protein-coupled receptor signalling pathway, Chemical synaptic transmission, and Heart contraction (S37). For this set of transcripts in males (3/5) we found GO terms related to protein folding (S40). Female transcripts with higher expression in FLXsex and FLXauto compared to CFMauto had GO terms related to ATPase activity (S40). Similar to group A, the pattern of differences in this group is consistent with a feminization in FLX genotypes compared to CFM autosomes.

(vii) Group G

We found 291 significant transcripts in group G in females and 29 in males (Table 1). Characteristic for this group is that FLXsex and CFMsex are significantly different from CFMauto (Figure 4G). Female transcripts with higher expression in CFMauto compared to the other genotypes (199/291) were enriched for genes associated with expression in the carcass and head (S31), as well as mito-annotated and mito-proteome genes (S43). There was one GO term associated with this set of transcripts; Generation of precursor metabolites and energy (S42). For this set of transcripts in males (19/29) we found enrichment of genes related to sexual conflict in *D. melanogaster* (S16). Female transcripts with higher expression in FLXsex and CFMsex than in CFMauto had one associated GO term; Egg chorion (S42). The pattern of differences in this group suggests that it is associated with masculinization of autosomes as a result of adaptation to the presence of the FM balancer.

4. Discussion

Here, we attempted to disentangle effects of evolved sex chromosomes and autosomes from a set of pre-existing experimental evolution lines. The experiments presented here were motivated by mounting evidence that the FM balancer which was used to enforce female-limited inheritance of the X-chromosomes in the FLX regime (and which was also present in the methodological control regime CFM) seemed to have altered selection pressures in unexpected ways [9–11]. Because the FM balancer carries markers that affect eye development, male carriers of balancer are blind. This seems likely to have affected the intensity and/or direction of sexual conflict in the FLX and CFM selection regimes, since sighted females that are heterozygous for the FM balancer can probably exert greater control over mating rates when paired with blind FM males compared to sighted wildtype males. We therefore hypothesized that the autosomes in the FM-bearing selection regimes may have evolved masculinization in order to compensate for increased female control over mating rates. This should result in a pattern of relative masculinization in genotypes with CFM and FLX autosomes compared to CFM and FLX sex chromosomes, while the female-limited selection in the X chromosome in the FLX selection regime should cause genotypes with FLX sex chromosomes to be feminized relative to those with CFM sex chromosomes. Our results were partly consistent with these predictions.

(a) Phenotypic assays

We have previously found evidence that CFM males were better at sperm offence than both FLX and Cwt males [10], so apart from the predictions above, we expected that males from one or both of the CFM genotypes (CFMauto and CFMsex) might perform better at sperm offence than the FLX genotypes (FLXauto and FLXsex). This proved not to be the case, since there were no significant differences in sperm offence between genotypes (S4B & S5B). We have also previously found evidence that FLX males were better at sperm defence than both CFM and Cwt males, and therefore similarly expected that one or both FLX genotypes might outperform the CFM genotypes in sperm defence, but there were again no significant differences (S4A & S5A). Why we did not detect any differences when the effects of the evolved sex chromosomes and autosomes were disentangled is unclear, but it suggests that evolved differences in sperm competitive ability cannot be attributed to a specific part of the genome.

Locomotion activity has previously been shown to be a sexually antagonistic trait in *Drosophila* [36], and we found a trend towards significant differences in locomotory activity between selection regimes [9]. In addition, other results from the experimental evolution lines suggest that the genetic architecture of locomotory activity has changed in the FLX regime (Manat *et al.* 2026, submitted). We therefore speculated that disentangling the effects of the evolved sex chromosomes and autosomes could result in clearer differences. This proved to be the case. Consistent with our predictions, females with FLX sex chromosomes had significantly lower locomotory activity than females with CFM sex chromosomes and CFM autosomes, indicating feminization of the FLX X chromosome (Figure 2, S5C&D). This suggests that our selection protocol did successfully result in feminization of the X in the FLX selection regime, but that this effect was partially cancelled out by adaptation on the autosomes for at least some traits [9].

(b) Expression data

Analysis of expression data provided some support for our *a priori* predictions, but also some unexpected patterns.

(i) *A priori* predictions

We predicted masculinization of expression in the FLX and CFM autosomal genotypes compared to the FLX and CFM sex chromosome genotypes. Support for this prediction was found in group A (the most numerous group, where CFMauto differed significantly from all the other genotypes, Figure 4A) and group G (where CFMauto differed significantly from both sex chromosome genotypes, Figure 4G). The pattern in group A was the most consistent with our *a priori* prediction (despite not detecting a difference between FLXauto and FLXsex) since transcripts that were more highly expressed in CFMauto were significantly associated with male fitness and male-benefit sexual antagonism.

We also predicted feminization of FLX X chromosomes compared to CFM X chromosomes, but in contrast to locomotory activity, there was little direct support for this prediction in the expression data. This is somewhat surprising, since we assumed that phenotypic differences would be reflected in differences in expression. However, if the differences in locomotory activity are due to changes in only a few genes, then this might make it difficult to detect a signal amongst all the other changes in expression we found. We also found patterns consistent with overall changes between the FLX and CFM selection regimes in groups E and F (Figure 4E&F), and although we did not detect a clear signal of feminization in these comparisons, results from groups E and F suggest that our selection regimes caused changes in genes that are associated with sexual conflict.

(ii) Unexpected patterns

Three groups revealed unexpected patterns that we did not predict. In group B (the second most numerous group, where CFMsex was significantly different from CFMauto, Figure 4B) we found evidence of feminization of CFM autosomes compared to CFM sex chromosomes. This runs counter to our prediction that adaptation to the presence of the FM balancer should have occurred primarily via the autosomes. However we found GO terms associated with RNA maintenance in this group, which suggests that part of the response we found could be related to compensation for loss-of-function mutations on the FM balancer. Expression differences in this group could also be associated with maintenance of male-specific selection on the X in the

CFM regime compared to the FLX regime, resulting in apparent masculinization compared to the other genotypes.

Secondly, in group C (where FLXauto was significantly different from CFMauto, Figure 4C), we found evidence of masculinization of FLX autosomes compared to CFM autosomes, which is another pattern that we did not expect *a priori*. However, these results could be consistent with predictions from Muralidhar & Coop [5], if autosomal genotypes evolved to partially counteract the effects of feminization of the X chromosome in the FLX selection regime.

Finally, we found evidence of feminization of CFM autosomes compared to FLX autosomes in group D (where CFMauto was significantly different from both CFMsex and FLXauto, Figure 4D). We speculate that changes in this group could potentially be due to mitonuclear effects, partly because of the fact that GO terms for this group were primarily associated with the mitochondria, but also because one of the main differences between the FLX and CFM selection regimes is that the mitochondria and FM balancer are co-inherited once every three generations in the CFM regime, but never co-inherited in the FLX selection regime in females. Any deleterious interactions between the FM balancer and mitochondrial genome could therefore evolve to be compensated for by the autosomes in the CFM selection regime.

(c) Overall patterns and general conclusions

In addition to the specific comparisons discussed above, we also found some more general results reflected in all groups, suggesting common themes of adaptation across our selection regimes. For example, genes associated with expression in ovaries were typically overrepresented in comparisons where expression was higher in FLX genotypes compared to CFM genotypes, consistent with feminization in the FLX selection regime. Interestingly, the head was also another tissue overrepresented among all seven groups, which is makes it a relevant tissue to follow up on specifically. Another general result was the overrepresentation of genes associated with sexual conflict in all comparisons, which fits well with previous results suggesting that greater investment in sexual conflict traits results in a shorter lifespan in CFM males [11]. Mitonuclear effects and/or selection on metabolism is another recurring theme across almost all groups. This is consistent with results from this and other species, which suggest that sex-specific selection on metabolism is an important component in sexual conflict [37–39]. Somewhat counterintuitively, given the original selection protocol and the composition of our target genotypes, X-linked genes were almost never overrepresented among our groups of transcripts. However, this pattern is consistent with our previous studies showing that selection on restricted parts of the genome can have genome-wide responses [37,38,40].

But arguably the most striking result was the fact that we detected an order of magnitude fewer significant transcripts in males than in females across all groups of comparisons. This was unexpected, since previous datasets from these and other experimental evolution studies have found similar numbers of significant transcripts between the sexes [41,37,38]. We have previously investigated how mismatching of the X and Y chromosomes affects male fitness, and found that males with mismatched X or Y chromosomes had higher fitness, but that males with matched X and Y chromosomes together with mismatched autosomes had similar fitness to wildtype males. We therefore concluded in that study that interactions between the X and Y are more important in determining male fitness than X-autosome or Y-autosome interactions [4]. However, we did not investigate the effects of mismatched X chromosomes and autosomes in females in that study, so the results presented here provide an interesting complement to this previous experiment [4]. Our male genotypes in this experiment had matched sex chromosomes (of FLX, CFM, or Cwt origin) that were paired with mismatched autosomes, and this had no significant effect on the phenotypic traits we measured, and only a modest effect on expression profile, consistent with previous results. The extensive effects on expression which we detected here when matched X chromosomes are paired with mismatched autosomes in females therefore suggests that X-autosome interaction effects are much more important in females than in males in *Drosophila melanogaster*.

In conclusion, we found definite evidence of sex-specific adaptation in different parts of the genome in our experimental genotypes, but the patterns were considerably more complex than we had initially expected. However, we believe that these results demonstrate that attempting to disentangle effects of X, Y, and autosomes is worthwhile to better understand sexually antagonistic intragenomic coevolution. Together with results from comparative studies, which have shown that genes that benefit one sex tend to accumulate in particular regions of the genome [6,42], our results highlight the fact that the genome is both dynamic and interconnected, resulting in sometimes surprising coevolutionary effects.

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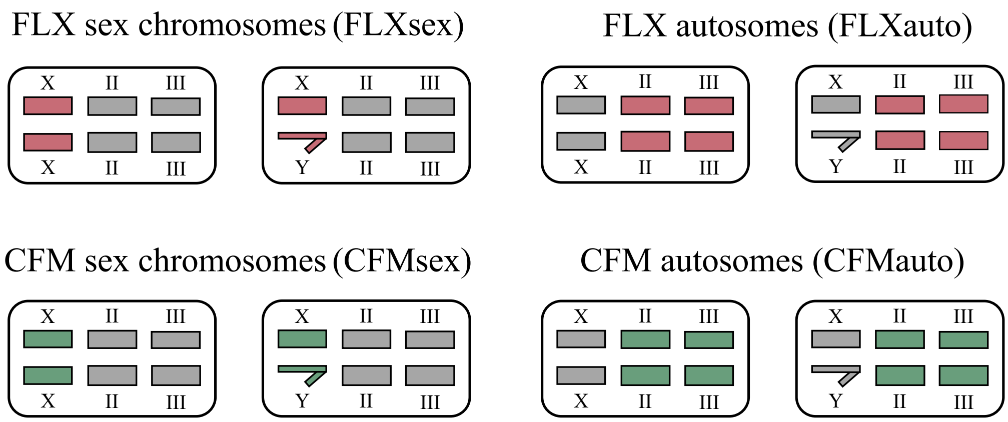

Figure 1: The four novel genotypes. Autosomes (II and III) and chromosome X are depicted as rectangles, and the chromosome Y as a half arrow. The small dot chromosome IV is not shown. "FLX sex chromosomes" (FLXsex) have sex chromosomes originating from the FLX selection regime (red chromosomes) and autosomes from the Cwt control regime (grey chromosomes). "FLX autosomes" (FLXauto) have sex chromosomes from Cwt (grey chromosomes) and autosomes from FLX (red chromosomes). "CFM sex chromosomes" (CFMsex) have sex chromosomes originating for the CFM control regime (green chromosomes) and autosomes from Cwt (grey chromosomes). "CFM autosomes" (CFMauto) have sex chromosomes from Cwt (grey chromosomes) and autosomes from CFM (green chromosomes).

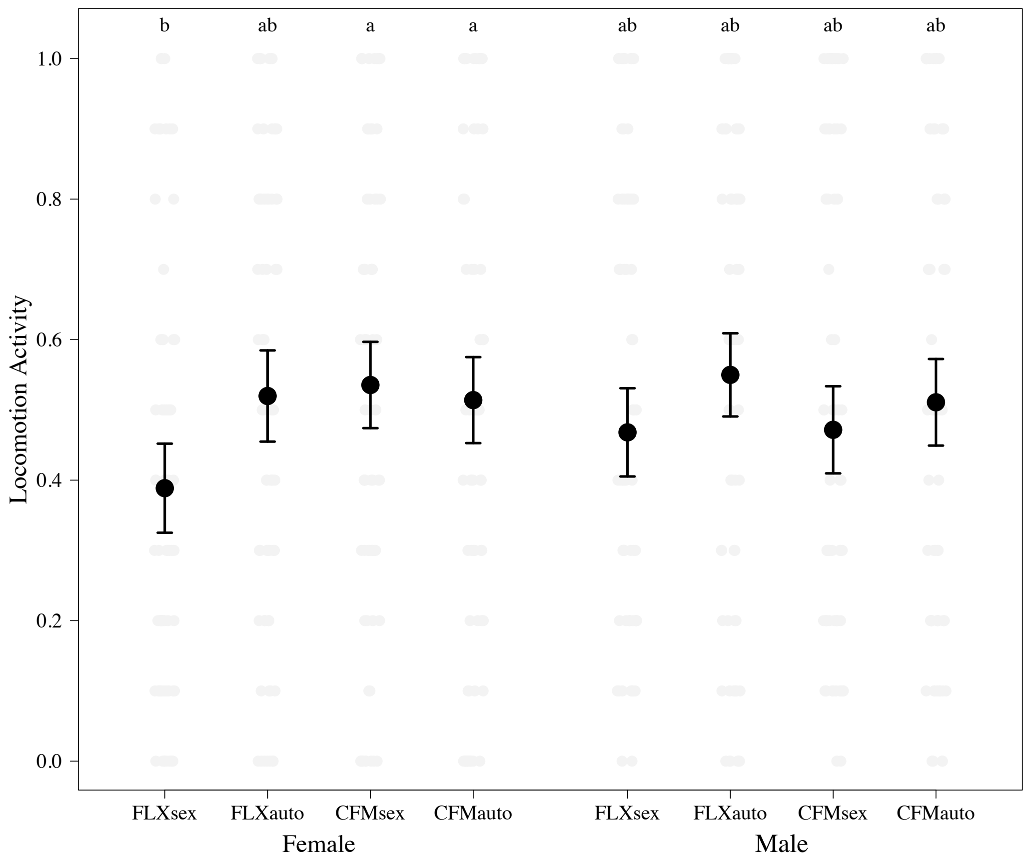

Figure 2: Locomotory activity of both sexes. The raw data is plotted as grey points and the black circles are the mean (±SE) of fitted values from the models. Letters indicate significances (Tukey HSD; $P < 0.05$).

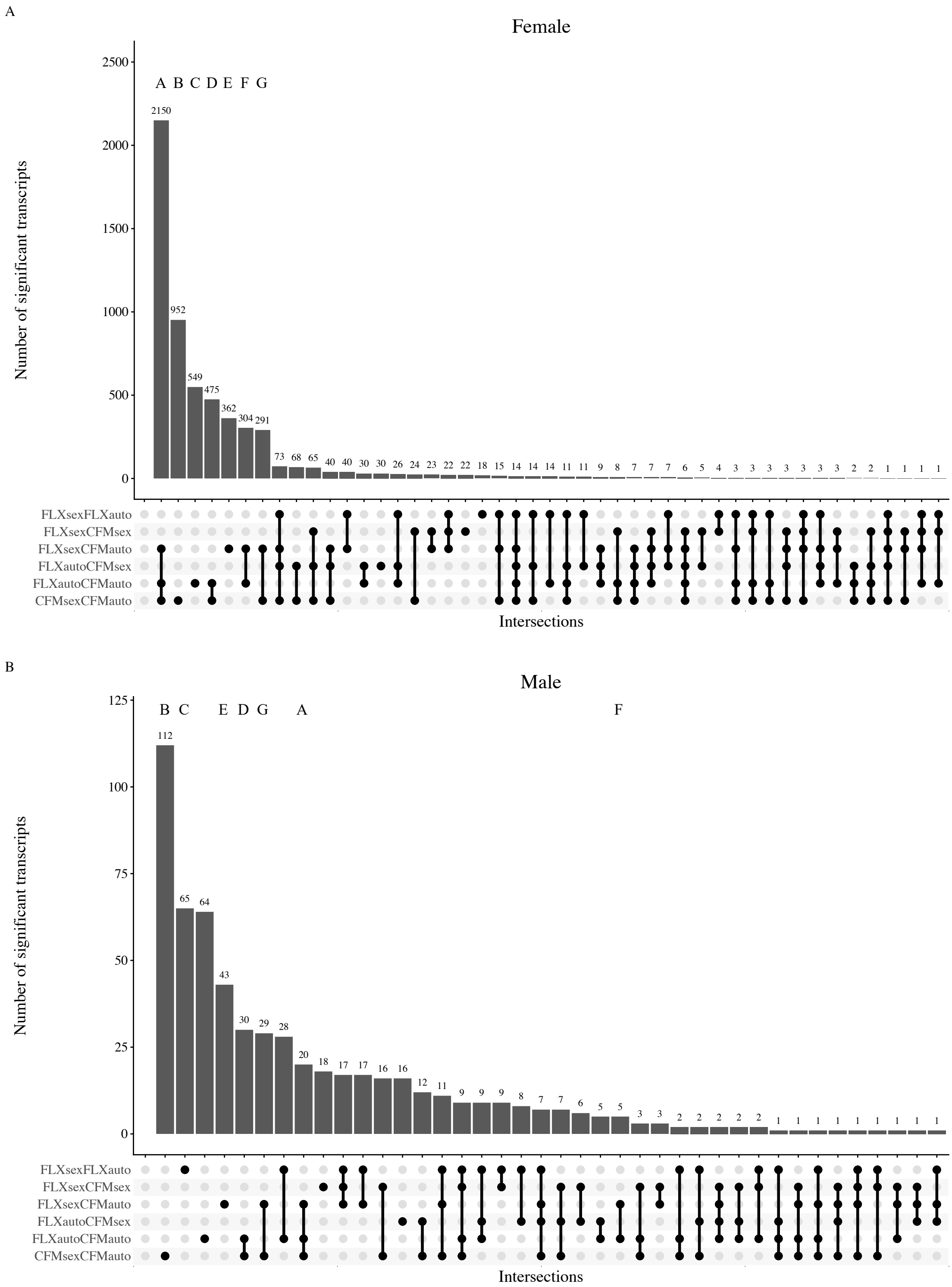

Figure 3: A) Female upset plot. There were 47 intersections with at least one significant differentially expressed transcript in the six pairwise comparisons between the four novel genotypes. The seven most numerous ones are labelled A-G in descending order, and are used in the downstream analysis. B) Male upset plot. There were 40 intersections with significant transcripts in males. The corresponding groups to the seven female groups are also labelled A-G.

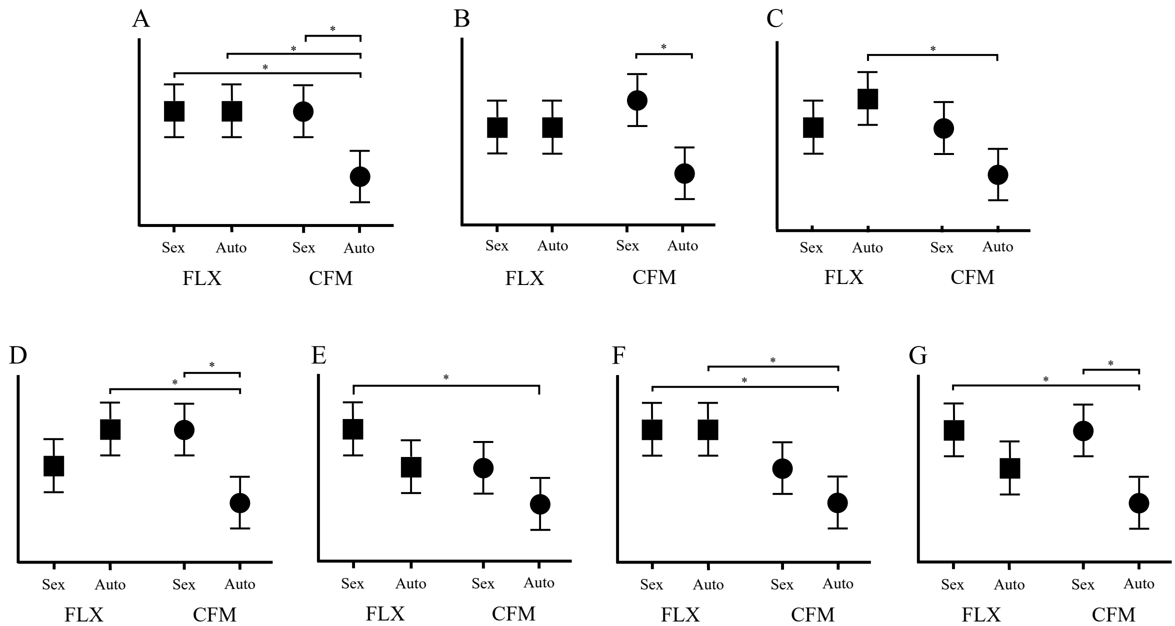

Figure 4: A schematic representation of the top seven most numerous groups of significant transcripts in the females. Patterns are representative of transcripts classified as upregulated; the pattern should be the reverse for downregulated transcripts.

Table 1: The top seven most numerous groups in females and the corresponding groups in males

Group	Sex	Total	Up-regulated	Down-regulated
A	Female	2150	129	2021
	Male	20	4	16
B	Female	952	570	382
	Male	112	54	58
C	Female	549	177	372
	Male	64	27	37
D	Female	475	191	284
	Male	30	14	16
E	Female	362	98	264
	Male	43	17	26
F	Female	304	40	264
	Male	5	2	3
G	Female	291	92	199
	Male	29	10	19